

Hard Disk Drive (HDD) & Partitioning your disk for Linux

As with all things Linux, it's possible to do it in the smallest of setups. Using distributions such as [Puppy Linux](#), you can achieve a fully working Linux setup in a few hundred megabytes. However, if you want to install a standard desktop installation of any up-to date distribution, you will probably want at least 20-40GB (gigabytes) free hard disk space. If you are going for the plunge and will convert your entire system over to Linux, then the more the better – 100GB+ in order to store all of your stuff: Apps, MP3s, Movies, Documents, emails etc and over time, it uses up quite a lot of drive space.

Modern Linux distributions easily support new drive technologies such as software RAID and SATA out of the box. Enterprise grade iSCSI or fibre channel disk arrays, are supported by distributions like Ubuntu Server edition or RHEL.

SSD (Solid State Drives) have become commonplace on higher end laptops and PCs these days and increase read and write speed significantly, making the whole system feel faster. This benefit is mirrored in Linux with an SSD.

Another option is to purchase a new hard disk to install Linux on or recycle an old hard drive if you have one spare! The reason for using a separate disk is because you are likely to be using another Operating System already such as Microsoft Windows or Mac OS. If you wish to use both Linux and Windows/Mac OS (so you can see if Linux is for you), then the easiest way to set it all up is if you have another drive to put Linux onto. You won't have to mess around with resizing partitions and the like.

- Minimum Specification: 5 GB (although some distributions can be smaller).
- Recommended Specification: Minimum of 25GB, or as much as you can afford to give Linux.

(Re)-Partitioning

Typically, Windows/Mac OS will allocate 100% of your computer's hard drive to it's own use, meaning there is no space left for Linux. If you don't want to buy a new hard drive for Linux, then you will somehow have to re-allocate some of the unused (free) space on your Windows/Mac OS drive for Linux. The act of slicing up the space on a hard drive into distinct segments is called Partitioning.

Thankfully, recent versions of popular Linux distributions now make it a snap to re-partition your disk. They work by utilising the free space that you have in your Windows drive (say your C: Drive) and creating a partition out of some or all of that free space for Linux. Although not necessary, you can also use something like the freely available [GParted](#) or commercially available [Acronis Disk Director](#) (for Windows). This allows you to split your disk into partitions as well as resize existing partitions before you *even start* to install Linux. It gives you absolute control of the whole process.

Give Linux as much breathing room as you can afford to!

If you are resizing your Windows partition to accommodate the installation of Linux, try and devote as much space to Linux as you can manage. If for example you have 100GB unused/free space on an 500GB drive, resize your windows partition down from 500GB to 410GB, leaving 90GB for linux and 10GB 'breathing room' spare for windows. This way you probably won't have too much concern about free disk space in the future.

Example: You can split it any way you like, here is an example of how your hard disk would look, if drawn as a sideways graph. This shows an example of roughly 30% Windows, 70% Linux. You would resize the Windows partition down to the size you want, which would give you the remainder of the disk as unallocated space. You can then use that unallocated space for Linux. Note in this example, most of the 70% Linux space is 'EXT4', however a small amount (say 8GB) is dedicated as 'SWAP'. SWAP is the area which if the memory fills up on your machine, it will use the disk instead. This is useful if you only have limited RAM (Memory), however on higher RAM specs, you probably won't need it.

The [GParted](#) tool is great for partitioning your hard drive. In this example you can see two Windows NTFS partitions (sda1 and sda2), then two linux partitions (sda5 and sda6).

Although the process of re-partitioning and dual-booting your PC with Linux and Windows is far easier than it used to be, to a computer novice it can still appear to be a daunting task. Don't worry though, I've got it all covered in [Chapter 5](#).